

ME594: 3D Robotics, HW 7

Homework due on 03-09-2025, Topics: Jacobian.

1. **Jacobian of the end-effector:** Consider the Unitree A1 quadruped leg shown in Fig. 1. The leg has three degrees of freedom: q_a , q_h , and q_k . The axes are as shown. The distance between first two joints is w as shown and the thigh and shank length are the same, L . The body frames when all the angles are 0 are shown. Also, the leg points vertically downward when all the angles are zero. However, the figure shows the configuration at some non-zero q_h and q_k . The world frame is at the same position as frame 1 when $q_a = 0$. The end-effector is the foot location, which is shown using a yellow sphere. Answer the questions below.

- (a) Write formula for H_1^0 , H_2^1 , H_3^2 in terms of L , w , q_a , q_h , q_k .
- (b) Compute the numerical value of Jv (size is 3×3 matrix) for the end-effector. Use these values $L = 0.2$ m, $w = 0.08505$ m, $q_a = 0.1$, $q_h = 0.3$, $q_k = -2.2$. Please indicate the formula used. This will help me grade your solution.

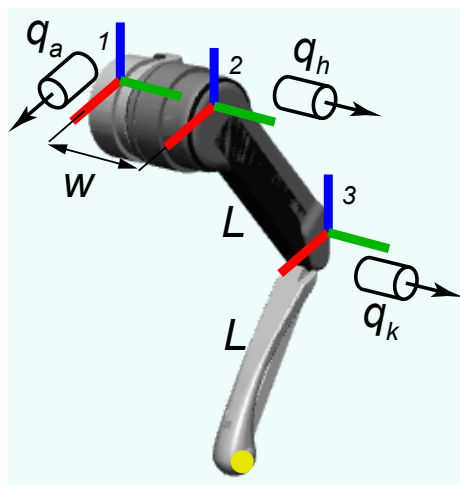


Figure 1: **Unitree A1 leg:** The body frames are given. The end-effector is shown with a yellow sphere. The goal is to compute the linear component of the Jacobian of the foot.

Submission:

- **Gradescope:** Submit solutions to both the parts.
- **Email:** None

2. **Jacobian for the Trossen Robotics WidowX 250 S:** Figure 2 shows the WidowX 250S by Trossen Robotics. A video of the robot in action is here: https://www.youtube.com/watch?v=l_10IH0JNxI. I have provided the wx250s.xml files that already has the sensor tag defined, the remaining scene and meshes can be downloaded here: https://github.com/google-deeppmind/mujoco_menagerie/blob/main/trossen_wx250s/wx250s.xml

Given the following position and velocity for the first six joints

$q = \text{np.array}([0.5123, -1.6532, 0.5439, -0.3432, -0.1125, 0.2342])$

$u = \text{np.array}([0.2291, -0.8602, -0.8068, -0.7110, 0.0689, 0.457029])$.

- Compute the jacobian $\mathbf{J}$ of the center of mass of the body called “lower_forearm_link”.
- Use the Jacobian $\mathbf{J}$ and angular velocity of the joint u to compute the linear and angular velocity of the center of mass of “lower_forearm_link”. You can cross-check this against HW6 Problem 2. Do you get the same answer?

HINT: For the second part, if the jacobian of the center of mass is $\mathbf{J}$ and the angular velocity of the joints is u , then linear/angular velocity of the center of mass is given by: $[v, \omega] = \mathbf{J}u$.

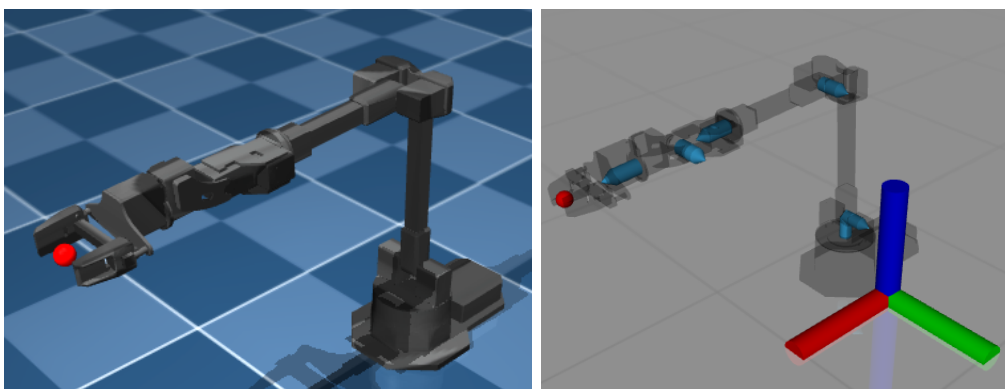


Figure 2: Perspective view of the Trossen WidowX 250 S: The robot has 6 joints as shown on the figure on the right. The world frame is at the base of the WidowX but moved to the side in the figure to avoid clutter. The body frames are in the same direction as the world frame and centered at the joints.

Submission:

- **Gradescope:** None.
- **Email:** Submit a zipped folder: *w250s_jacobian_names.zip* where *names* should be replaced by names of the team member(s) and sent to pranav@uic.edu. Please add “HW7 wx250s” to your subject line. If submitting as a group, please copy your group member.